

Summary of Qualifications

- Founder, innovation strategist, and leader in science and technology, bringing visionary systems thinking, relational intelligence, and strategic storytelling to complex technical and organizational challenges.
- 15+ years leading computational research, probabilistic modeling, machine learning, and data science to solve complex scientific and engineering challenges across frontier tech, academia, government, and startups.

Education

2015	Ph.D. in Astrophysics (Dissertation)	Pennsylvania State University, University Park
2012	M.S. in Astronomy (Master's Thesis Publication)	University of Florida, Gainesville
2009	B.S. in Physics (Senior Honors Thesis Publication)	University of California, Santa Cruz

Professional Experience

- 9/2025 - Present** **WOVEN - Riffit Woven Exchange** *Founder*
- Founded WOVEN, a globally sanctioned space for emergent cross-boundary teams to tackle high-stakes scientific and technological bottlenecks.
 - Defined the company vision, strategic positioning, and preliminary operating model for a new coordination layer connecting technical challenges with interdisciplinary contributors.
 - Designed frameworks for AI-supported facilitation, secure knowledge capture, and collaborative problem-solving to support flow and breakthrough insight generation.
 - Developed business model hypotheses around shared ownership and novel exchange mechanisms for collaboratively generated technical outputs.
 - Conducted founder-led discovery across deep-tech, aerospace, robotics, governance, investment, and research communities to assess future pilot pathways, strategic partnerships, and unmet needs in innovation infrastructure.

10/2022 - 12/2024 **Giant Magellan Telescope** *Lead Data Scientist and Solution Architect for Systems Engineering*

Independent Contributions

- Pioneered and built the Integrated Model Uncertainty Factor (IMUF) formalism and sensitivity analysis for systems engineering to quantify differences in telescope designs, contributing to organizational policy risk identification and mitigation.
- Removed human-in-the-loop error by increasing automation of the end-to-end integrated modeling pipeline.
- Migrated and modernized legacy computational fluid dynamics simulations and finite element analyses to the cloud using AWS ParallelCluster.
- Optimized the cost and performance of high-performance computing (HPC) infrastructure on AWS, achieving a 5x reduction in runtime and cost by identifying and deploying the x2iedn.8xlarge instance for finite element analysis.
- Developed a Sankey-based data visualization concept to map systems engineering flow-down requirements and budget allocations.
- Reported on and provided statistical analysis to observatory subsystems for simulated telescope performance metrics over a standard year of telemetry data taken from the telescope build site in the Atacama desert of Chile.

Leadership

- Appointed ground-based telescope community representative on NASA's Habitable Worlds Observatory (HWO) AI/ML Steering Committee.
- Formed NASA's HWO Generative AI task group, bringing together 19 professionals from industry, academia, government, multiple NASA centers, the Space Telescope Science Institute, and Northrop Grumman ([Link to white paper](#)).
- Directed a proposal to integrate generative AI into GMT's documentation and requirements processes; the proposal was accepted into the AWS Generative AI Innovation Center.
- Identified new opportunities for GMT's funding strategy and social-impact messaging related to AI integration.
- Served as a technical culture liaison at the AWS + GMT Executive Briefing Summit and AWS Think Big Research Forum.
- Led a global team of 11 to develop a GMT-enabled solution addressing a longstanding blocker in astronomy related to the detection and characterization of Earth twins.
- Established a collaboration with the Right Question Institute to apply structured inquiry methods to research team cohesion.
- Developed and presented "Generative AI LLMs in Support of Collective Creativity for Open Science" ([Link to presentation recording](#)).
- Uncovered critical Earth-analog exoplanet science that can only be accomplished with the unique capabilities of the GMT.

4/2019 - 2/2022 Bountiful (formerly Vinsight YC W17) Head of Data Science / Lead Machine Learning Engineer
Managerial Functions

- Led a hybrid data science and machine learning team of 3, reporting to the Head of Product and CEO.
- Developed and executed the company data science vision and roadmap in partnership with product leadership.
- Contributed to growth across key business metrics, including \$2M in marketplace transactions, 70K acres of customer field-level data, 136 monthly active users, and 27 weekly active users.
- Fostered psychological safety and professional growth that enabled the team to overcome key technical hurdles.
- Authored company blog posts [1,2] on the generative model yield-forecasting innovations I led; for a period, they ranked as the second top Google search result for "Bayesian Data Analysis."
- Supported \$3M in seed fundraising.

Independent Contributions

- Improved decision making for agriculture by developing multilevel forecasting models for specialty crop yield, outperforming legacy machine learning models by 30%.
- Scaled models to accommodate a 2X increase in customers via parallelization techniques, improving runtime for all customer data from 7hrs. to 45 min.
- Utilized Google Earth Engine, satellite data including Landsat, weather stations, 5-7 years of historic data from customers, and phenological data for specialty crops.
- Provided robust specialty crop supply estimates to use for nitrogen allocation and application scheduling to prevent over application and conserve estuaries, saving farmers money throughout the growing season.
- Created market insights for specialty crop sales by rapidly prototyping and validating demand forecasting models.
- Received mentoring from [Bountiful adviser](#) Hal Varian, the Chief Economist at Google.

1/2016 - 1/2019 NASA's Kepler Space Science Telescope Mission NASA Postdoctoral Program Prize Fellow

- Innovated a statistical approach that included the possibility that some exoplanet radius measurements detected by Kepler were, in fact, Earth-twins.
- Constructed statistical frameworks for exoplanet occurrence rates incorporating the uncertainty in planet radius using hierarchical Bayesian generative models and Hamiltonian Monte Carlo for data products with ~ 5000 planet candidates.
- Worked with the Kepler mission team on the development and application of exoplanet detection reliability metrics.
- Conducted speckle interferometric observations at Gemini South that identified a bias in exoplanet radii measurements that revealed unresolved double stars.
- Evaluated technical proposals for data collection with NASA's Kepler 2 space mission, and applications for the NASA Graduate Research Fellowship Program.
- Acted as Data Science Instructor for the Large Synoptic Survey Telescope Corporation Data Science Fellowship Program at the University of Washington eScience Institute, teaching 60+ graduate students ([github repo](#)).
- Invited contributor to the 2018 NASA Technosignatures Workshop, which convened 50 invited contributors and 4,000+ remote viewers and produced four technical reports to NASA ([Link to report document](#)).
- Organizer for StanCon2018 with 200+ participants from industry and academia over a period of 3 days.
- Presented science objectives and results from NASA's Kepler Mission on behalf of NASA international partnerships to NASA business partners.
- Invited scientist for the 2018 Australian National Science Week (presentations to audiences of 600+, radio shows, podcasts, and TV interviews).

8/2013 - 12/2015 Penn State Center for Exoplanets and Habitable Worlds Graduate Student Researcher

- Utilized hierarchical Bayesian statistical analysis to understand exoplanet formation using Python, R, and JAGS.
- Published [10+ refereed papers](#) in The Astrophysical Journal and Monthly Notices of the Royal Astronomical Society.
- Presented at 13 conferences, including 9 talks and 4 poster sessions, at events such as the American Astronomical Society annual meeting (3000+ attendees) and the International Society for Bayesian Statistics (750+ attendees).
- Acted as assistant teacher for researchers attending the High-Performance Computing for Astronomical Data Analysis Summer School at the Penn State Center for Astrostatistics.

8/2010 - 7/2013 University of Florida, Astronomy Dept. Graduate Student Researcher

- Built Python analysis tools and radiation hydrodynamics algorithms to study exoplanet moon formation and the mass evolution of circumplanetary disks.
- Contributed to the analysis of the first ALMA radio and infrared telescope data to characterize Fomalhaut, a planetary debris disk.
- Planned and led an introductory astronomy lab for 60 undergraduates.

1/2009 - 7/2010 University of California, Santa Cruz, Astronomy and Astrophysics Dept. Junior Specialist

- Evolved exoplanet atmosphere code to investigate non-equilibrium chemistry and 3D atmosphere profiles.
- Tasked with training undergraduate astrophysics majors in advanced calibration routines, spectroscopy, photometry and radio astronomy.
- Mentored students with laboratory set-up and demonstrations for high school AP physics and conceptual physics classes.

Skills

Founder and Executive Leadership: Vision development, strategic storytelling, cross-functional leadership, stakeholder alignment, ecosystem and community building, public speaking, media engagement

Strategy and Operations: Roadmapping, budgeting, partnership development, proposal and grant strategy, technical risk analysis, AI/ML strategy, program leadership

Scientific Computing: Statistical modeling, machine learning, high-performance computing, data governance, reproducible research, peer and panel review, journal publications

Programming and Infrastructure: Python, R, Rust, SQL, Stan, JAGS, C, Fortran, Git, Docker, AWS (ParallelCluster, Batch, SageMaker, FSx, RDS, CloudFormation), Google Earth Engine

Technical Program and Collaboration Tools: JIRA, Microsoft SharePoint, IBM DOORS, SolidWorks Manage

Awards and Qualifications

- 2016** NASA Postdoctoral Program Prize Fellowship at NASA Ames Research Center
- 2013** Homer F. Braddock Scholarship in Biology, Chemistry, and Physics at Penn State University
- 2010** Grinter Fellowship at University of Florida
- 2010** Schrader Fellowship at University of Florida
- 2008** Cal Teach Award at University of California, Santa Cruz

Volunteering

- 2025-Present** Co-Host for weekly SF/Stanford Intellectual Salon
- 2024** NASA Habitable Worlds Observatory AI/ML Steering Committee Member and Gen AI Task Group Leader
- 2019-2022** American Astronomical Society Sustainability Committee
- 2019-2022** GetSET Society Of Women Engineers Santa Clara County